

Introduction to Cell Biology

Chapter 1: The Cell

All living organisms are composed of cells, the basic structural and functional unit of life. Cells were first observed by Robert Hooke in 1665, and the cell theory, formalized in the 1830s, states that all living things are made of cells, the cell is the basic unit of life, and all cells arise from pre-existing cells.

There are two broad categories of cells: prokaryotic and eukaryotic. Prokaryotic cells, found in bacteria and archaea, lack a membrane-bound nucleus and most organelles. Eukaryotic cells, found in plants, animals, fungi, and protists, contain a true nucleus and a variety of specialized organelles.

1.1 The Cell Membrane

The cell membrane, also called the plasma membrane, is a selectively permeable barrier that separates the cell's interior from its external environment. It is composed primarily of a phospholipid bilayer embedded with proteins, a structure described by the fluid mosaic model.

The membrane controls what enters and exits the cell through passive transport (diffusion, osmosis) and active transport, which requires energy in the form of ATP to move substances against their concentration gradient.

1.2 The Nucleus

The nucleus is the control center of the eukaryotic cell, housing the cell's DNA organized into chromosomes. It is surrounded by a double membrane called the nuclear envelope, which contains pores that regulate the movement of molecules between the nucleus and the cytoplasm.

Within the nucleus, the nucleolus is responsible for producing ribosomes, which are later transported to the cytoplasm to carry out protein synthesis.

Chapter 2: Cellular Respiration

Cellular respiration is the process by which cells convert nutrients, primarily glucose, into usable energy in the form of ATP (adenosine triphosphate). This process occurs mainly within the mitochondria, often called the powerhouse of the cell.

Cellular respiration consists of three main stages: glycolysis, which occurs in the cytoplasm and produces a small amount of ATP; the Krebs cycle, which occurs in the mitochondrial matrix; and the electron transport chain, which produces the majority of the cell's ATP.

2.1 Glycolysis

Glycolysis is the first stage of cellular respiration and does not require oxygen, making it an anaerobic process. During glycolysis, one molecule of glucose is broken down into two molecules of pyruvate, yielding a net gain of 2 ATP and 2 NADH molecules.

2.2 The Electron Transport Chain

The electron transport chain is located in the inner mitochondrial membrane and consists of a series of protein complexes that pass electrons from NADH and FADH₂ down an energy gradient. This process drives the production of a large majority of the cell's ATP through a process called chemiosmosis, and requires oxygen as the final electron acceptor, making it an aerobic process.

Chapter 3: Cell Division

Cell division allows organisms to grow, repair damaged tissue, and reproduce. In eukaryotic cells, division occurs through one of two processes: mitosis, which produces two genetically identical daughter cells, or meiosis, which produces four genetically distinct gametes with half the chromosome number of the parent cell.

Mitosis consists of four main phases: prophase, metaphase, anaphase, and telophase, collectively known as the mitotic phase, which is preceded by interphase, during which the cell grows and replicates its DNA.